ADITYA UNIVERSITY

SCHOOL OF COMPUTING

Department of Artificial Intelligence & Machine Learning

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"***STUDENT BUDGET ONLINE SHOPPING SYSTEM***"* is submitted by

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In partial fulfilment of the requirements for the award of the B.Tech. degree in Artificial Intelligence & Machine Learning for the academic year 2026-2027.

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STUDENT BUDGET ONLINE SHOPPING SYSTEM

A DBMS Capstone Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted By

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

Aditya Nagar, ADB Road, Surampalem, Andhra Pradesh, India

2026 – 2027

ADITYA UNIVERSITY
(Formerly Aditya Engineering College(A))
Aditya Nagar, ADB Road, Surampalem, Andhra Pradesh, India
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



CERTIFICATE

This is to certify that the DBMS Capstone Project report entitled "Student Budget Online Shopping System" is being submitted by ADARTA SUMALATHA (26B21AI001), Y. YAMUNA MANASA VIHARI (26B21AI116), K. SRAVANI JYOTHI (26B21AI059), R. SRUTHI ALEKHYA (26B21AIA06) partial fulfillment of the requirements for the award of the B.Tech. degree in Artificial Intelligence and Machine Learning. The work documented in this report was carried out as part of the **Capstone Project** for the course **Database Management Systems (2501IT05)** during the academic year **2026–2027**, and records the database design, implementation, and documentation undertaken by the team.

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DECLARATION

We hereby declare that the DBMS Capstone Project entitled "Student Budget Online Shopping System" is a genuine work carried out by us. The database design, SQL implementation, queries and documentation presented in this report are our own work, completed under the guidance of Mrs. G. L. Narasamba, Department of Artificial Intelligence and Machine Learning, Aditya University, Surampalem. We further declare that this work has not been submitted, in full or in part, for the award of any degree of this or any other educational institution.

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Date: 14 september 2026
Place: Surampalem

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We extend our sincere thanks to the Management, Dr. M. Sreenivasa Reddy, Deputy Pro-Chancellor, Dr. M. B. Srinivas, Vice-Chancellor, and Dr. A. Ramesh, Pro Vice-Chancellor (Engineering & Sciences), for providing an excellent academic environment.

Not to forget, we thank the Teaching and Non-Teaching Staff, our parents and our friends, who have directly or indirectly supported us in completing this project on time.

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DBMS CAPSTONE PROJECT

Project Documentation Report

Project Title : Student Budget Online Shopping System

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Academic Year: 2026–27 Date of Submission: 14 September 2026

GitHub Repository Link: https://github.com/sravani-kotha/student-budget-online-shopping

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ABSTRACT

The **Student Budget Online Shopping System** is a database application designed to help students shop for essential items while staying within a limited budget. It stores information about students, budgets, product categories, products, shopping carts, orders, payments, wishlists and feedback in a centralized relational database.The database consists of eleven related tables: Student, Category, Product, Budget, Cart, Cart_Item, Order_Details, Order_Item, Payment, Wishlist and Feedback. Primary keys identify every record uniquely, and foreign keys connect dependent records to their parents, which preserves referential integrity. NOT NULL, UNIQUE, CHECK and DEFAULT constraints keep the stored data valid. The design is normalized up to Third Normal Form (3NF) to reduce redundancy and update anomalies.

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1. INTRODUCTION

1.1 Brief Information about the Project

The Student Budget Online Shopping System is a relational database application that lets students plan a spending budget and buy the things they need, such as stationery, books, electronics accessories and hostel essentials. Every activity, from setting a budget to giving feedback on a purchased product, is recorded in related tables, so spending can be compared with the planned budget at any time.

The project replaces scattered notes, bills and spreadsheets with one structured database that supports consistent storage and quick retrieval using SQL.

1.2 Purpose of the Database

The purpose of the database is to **efficiently manage student budgets and online shopping activities** in one centralized system. It stores and organizes information about students, products, categories, budgets, carts, orders, payments, wishlists, and feedback. It also helps track purchases and spending while maintaining **accurate, consistent, and organized records** using database keys and constraints.

2. PROBLEM STATEMENT

Students have **limited budgets** and need to control their spending.

Manual tracking of **prices, quantities, and purchases** is difficult.

A centralized DBMS is needed to manage **students, budgets, products, carts, orders, payments, and feedback** efficiently.

3. OBJECTIVES

- Objective 1: **Manage student budgets** and track spending.
- Objective 2:. **Store product details** such as price, stock, and category.
- Objective 3: **Manage shopping activities** like cart, wishlist, and orders.
- Objective 4. **Record payments and feedback** securely.
- Objective 5: **Maintain accurate and consistent data** using keys,constraints&SQL operatio

5.TOOLS & TECHNOLOGIES USED

Category	Tool / Technology Used
Database Server	SQ-compatible relational database
Query / Admin Tool	SQL client / SQL environment
Front-end (if any)	Not specified in the uploaded SQL files
Operating System	Not specified in the uploaded files
Version Control	GitHub

6.DATABASE DESIGN

6.1 Identified Entities

S. No	Entity Name	Description
1	**Student**	Stores student details such as name, email, phone number, department, and year.
2	**Product**	Stores product information including name, price, stock, and category.
3	**Budget**	Maintains the budget amount and spending period of each student.
4	**Order Details**	Records order date, total amount, and the student who placed the order.
5	**Payment**	Stores payment details such as payment date, amount, and payment method
6	**Cart**	Stores the products added by a student for shopping, along with cart details and creation date.

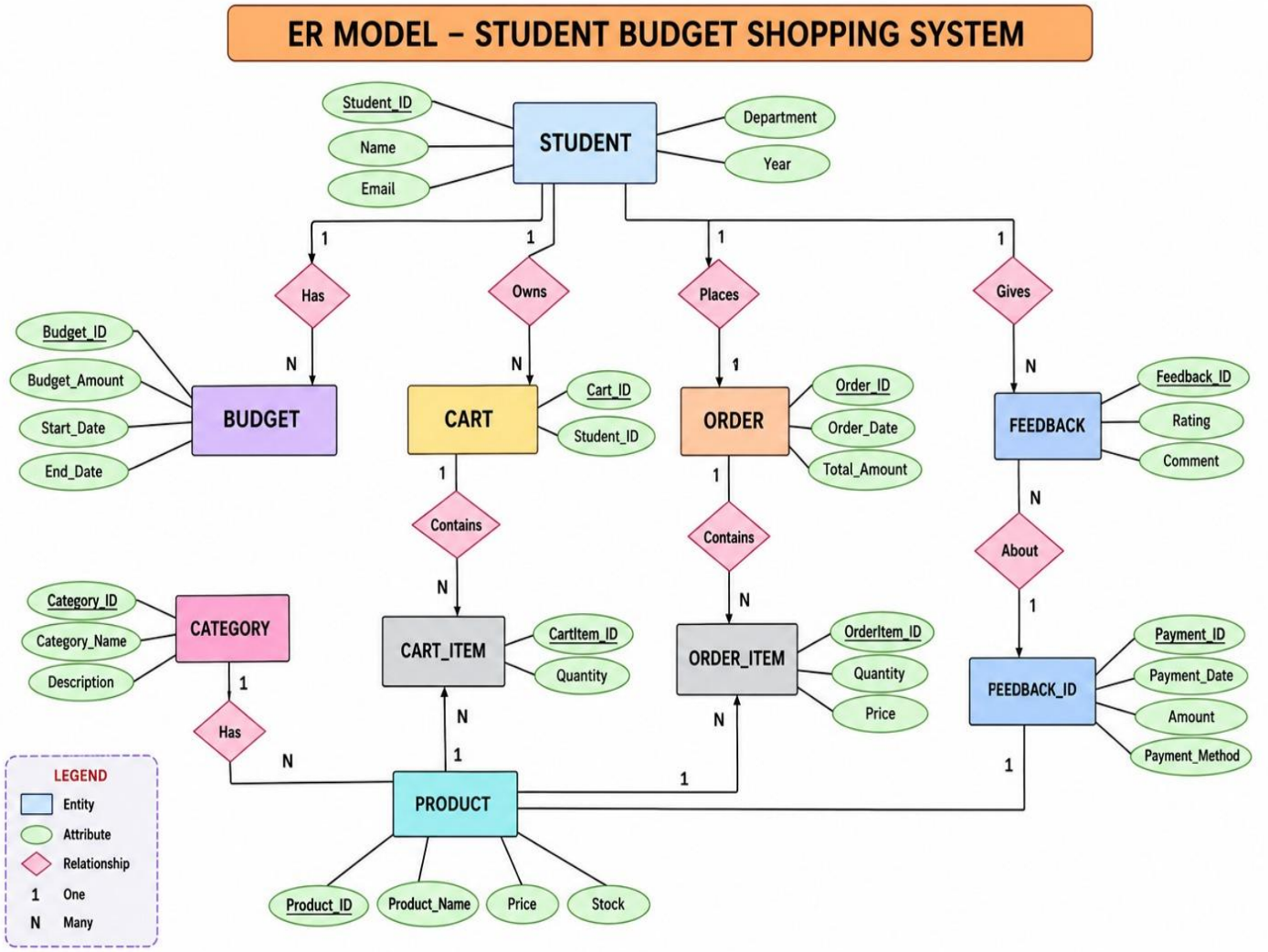
6.2 Entity Attributes

Entity	Attribute	Data Type	Key (PK/FK/—)	Attribute Type
Student	Student_ID	INTEGER	PK	Simple
Student	Phone_Number	VARCHAR(11)	—	Simple
Cart	City	VARCHAR(50)	—	Simple
Order Details	Total_Amount	DECIMAL(10,2)	—	Simple
Payment	Payment_Date	VARCHAR(100)	—	Simple
Product	Product_Name	DECIMAL(10,2)	—	Derived/Default
Product	Stock	INTEGER		Simple
Product	Category_ID	INTEGER	FK	Simple
Product	Price	DECIMAL(10,2)	—	Simple
Cart	Cart_ID	INTEGER	PK	Simple
Budget	Budget_ID	INETGER	PK	Simple

Budget	Budget_Amount	DECIMAL(10,2)	—	Simple
Budget	Student_ID	INTEGER	FK	Simple

Entity	Attribute	Data Type	Key (PK/FK/—)	Attribute Type
Customer	PANNo	VARCHAR(10)	—	Simple
Customer	Email	VARCHAR(100)	—	Simple
Customer	Phone	VARCHAR(12)	—	Simple
Customer	Address	VARCHAR(100)	—	Simple
Customer	CreatedAt	DATE	—	Derived/Default
Employee	EmployeeID	INTEGER	PK	Simple
Employee	BranchID	INTEGER	FK	Simple
Employee	FirstName	VARCHAR(30)	—	Simple
Employee	MiddleName	VARCHAR(30)	—	Simple
Employee	LastName	VARCHAR(30)	—	Simple
Employee	Designation	VARCHAR(50)	—	Simple
Employee	Qualifications	TEXT	—	Simple
Employee	Salary	DECIMAL(12,2)	—	Simple
Employee	Phone	VARCHAR(12)	—	Simple
Employee	Email	VARCHAR(100)	—	Simple
Employee	HireDate	DATE	—	Simple
Account	AccountNumber	VARCHAR(20)	PK	Simple
Account	CustomerID	INTEGER	FK	Simple
Account	BranchID	INTEGER	FK	Simple
Account	AccountType	VARCHAR(20)	—	Simple
Account	Balance	DECIMAL(15,2)	—	Simple
Account	OpenDate	DATE	—	Simple
Account	Status	VARCHAR(15)	—	Simple
TransactionRecord	TransactionID	INTEGER	PK	Simple
TransactionRecord	AccountNumber	VARCHAR(20)	FK	Simple
TransactionRecord	TransactionType	VARCHAR(15)	—	Simple
TransactionRecord	Amount	DECIMAL(15,2)	—	Simple
TransactionRecord	TransactionDate	DATE	—	Derived/Default
TransactionRecord	Mode	VARCHAR(20)	—	Simple
TransactionRecord	Description	VARCHAR(255)	—	Simple
Loan	LoanID	INTEGER	PK	Simple
Loan	CustomerID	INTEGER	FK	Simple
Loan	BranchID	INTEGER	FK	Simple
Loan	LoanType	VARCHAR(50)	—	Simple
Loan	LoanAmount	DECIMAL(15,2)	—	Simple
Loan	PrincipalAmount	DECIMAL(15,2)	—	Simple
Loan	InterestRate	DECIMAL(5,2)	—	Simple
Loan	StartDate	DATE	—	Simple
Loan	DurationMonths	INTEGER	—	Simple
Loan	EmiAmount	DECIMAL(10,2)	—	Simple
Loan	Status	VARCHAR(20)	—	Simple
Loan	SanctionDate	DATE	—	Simple

6.2 ER Diagram



6.4 Relationships & Cardinality

Entity 1	Relationship	Entity 2	Cardinality (1:1 / 1:N / M:N)
Branch	employs	Employee	1:N
Branch	has	Account	1:N
Branch	handles	Loan	1:N
Customer	owns	Account	1:N
Customer	takes	Loan	1:N
Account	has	TransactionRecord	1:N

6.5 Relational Database Schema

Table Name	Column Name	Data Type	PK/FK/—	Constraints
Branch	BranchID	INTEGER	PK	NOT NULL, FOREIGN KEY → Branch
Branch	BranchName	VARCHAR(50)	—	NOT NULL
Branch	IFSCCode	VARCHAR(11)	—	NOT NULL, UNIQUE, CHECK length=11
Branch	City	VARCHAR(50)	—	NOT NULL
Branch	State	VARCHAR(50)	—	NOT NULL
Branch	ManagerName	VARCHAR(50)	—	NOT NULL
Branch	CreatedAt	DATE	—	DEFAULT CURRENT_DATE
Customer	CustomerID	INTEGER	PK	NOT NULL, FOREIGN KEY → Customer
Customer	FirstName	VARCHAR(30)	—	NOT NULL
Customer	MiddleName	VARCHAR(30)	—	NULL allowed
Customer	LastName	VARCHAR(30)	—	NOT NULL
Customer	DOB	DATE	—	NOT NULL
Customer	Gender	VARCHAR(10)	—	CHECK Male/Female/Other
Customer	AadhaarNo	VARCHAR(12)	—	NOT NULL, UNIQUE, CHECK length=12
Customer	PANNo	VARCHAR(10)	—	NOT NULL, UNIQUE, CHECK length=10
Customer	Email	VARCHAR(100)	—	NOT NULL, UNIQUE
Customer	Phone	VARCHAR(12)	—	NOT NULL
Customer	Address	VARCHAR(100)	—	NOT NULL
Customer	CreatedAt	DATE	—	DEFAULT CURRENT_DATE
Employee	EmployeeID	INTEGER	PK	PRIMARY KEY
Employee	BranchID	INTEGER	FK	NOT NULL, FOREIGN KEY → Branch
Employee	FirstName	VARCHAR(30)	—	NOT NULL
Employee	MiddleName	VARCHAR(30)	—	NULL allowed
Employee	LastName	VARCHAR(30)	—	NOT NULL
Employee	Designation	VARCHAR(50)	—	NOT NULL
Employee	Qualifications	TEXT	—	NULL allowed

Table Name	Column Name	Data Type	PK/FK/—	Constraints
Student	Student_ID	INT	PK	NOT NULL, UNIQUE
Student	Name	VARCHAR(50)	—	NOT NULL
Student	Email	VARCHAR(100)	—	NOT NULL, UNIQUE
Student	Phone_Number	VARCHAR(15)	—	NOT NULL
Product	AccountNumber	VARCHAR(20)	PK	UNIQUE
Product	Product_Name	VARCHAR(100)	FK	NOT NULL,
Product	Price	DECIMAL(10,2)	FK	CHECK (Price > 0)
Product	Stock	INT	—	CHECK (Stock >= 0)
Product	Category_ID	INT	—	NOT NULL, REFERENCES
Budget	Budget_ID	INT	PK	NOT NULL,UNIQUE
Budget	Start_Date	DATE	—	NOT NULL
Budget	End_Date	DATE	—	NOT NULL
Cart	CART_ID	INT	PK	NOT NULL, UNIQUE
Cart	Student_ID	INT	FK	NOT NULL, REFERNCES Student
Cart	Created_Date	DATE	—	NOT NULL,
Cart_Item	Quantity	INT	—	CHECK (Quantity > 0)
Cart_Item	Product_ID	INT	FK	NOT NULL, REFERENCES Product
Cart_Item	Cart_ID	INT	—	NOT NULL REFERENCES Cart
Order_Details	Order_ID	INT	PK	NOT NULL,UNIQUE
Order_Details	Order_Date	DATE		NOT NULL, \
Order_Details	Student_ID	INT	—	NOT NULL, REFERENCES Student
Order_Item	Quantity	INT	—	NOT NULL, CHECK > 0
Order_Item	Price	DECIMAL(10,2)	—	CHECK > 0
Order_Item	Order_ID	INT	FK	NOT NULL, REFERENCES Product
Order_Item	Product_ID	INT	FK	NOT NULL
Wishlist	Added_Date	DATE	—	NOT NULL,
Feedback	Feedback_ID	INT	PK	NOT NULL, UNIQUE
Feedback	Rating	INT	—	CHECK (Rating BETWEEN 1 AND 5)

6.6 Normalization

1NF → 2NF → 3NF

1. **1NF (First Normal Form)**
   - Each column should contain only one value**.**
   - No repeating groups or multiple values in one cell.
   - Example: Store each product separately instead of putting many products in one column.
2. **2NF (Second Normal Form)**
   - Table must already be in 1NF**.**
   - Remove partial dependency.
   - Every non-key attribute should depend on the complete primary key**.**
3. **3NF (Third Normal Form)**
   - Table must already be in 2NF.
   - Remove transitive dependency**.**
   - Non-key attributes should depend only on the primary key**.**

7 DATABASE IMPLEMENTATION

7.1 DDL — Database and Table Creation

```
        ---- Drop tables if they exist
        DROP TABLE IF EXISTS Order_Coupon;
        DROP TABLE IF EXISTS Delivery;
        DROP TABLE IF EXISTS Coupon;
        DROP TABLE IF EXISTS Feedback;
        DROP TABLE IF EXISTS Wishlist;
        DROP TABLE IF EXISTS Payment;
        DROP TABLE IF EXISTS Order_Item;
        DROP TABLE IF EXISTS Order_Details;
        DROP TABLE IF EXISTS Cart_Item;
        DROP TABLE IF EXISTS Cart;
        DROP TABLE IF EXISTS Budget;
        DROP TABLE IF EXISTS Product;
        DROP TABLE IF EXISTS Category;
        DROP TABLE IF EXISTS Address;
        DROP TABLE IF EXISTS Student;
       -- 1. STUDENT TABLE
        CREATE TABLE Student (
            Student_ID INTEGER PRIMARY KEY,
            Name VARCHAR(50) NOT NULL,
            Email VARCHAR(100) NOT NULL UNIQUE,
            Phone_Number VARCHAR(15) UNIQUE,
            Department VARCHAR(50) NOT NULL,
            Year INTEGER CHECK (Year BETWEEN 1 AND 4));
         -- 2. CATEGORY TABLE
        CREATE TABLE Category (
            Category_ID INTEGER PRIMARY KEY,
            Category_Name VARCHAR(50) NOT NULL UNIQUE,
            Description VARCHAR(200));
        -- 3. PRODUCT TABLE
        CREATE TABLE Product (
            Product_ID INTEGER PRIMARY KEY,
            Product_Name VARCHAR(100) NOT NULL,
            Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
```

```
            Stock INTEGER NOT NULL CHECK (Stock >= 0),
            Category_ID INTEGER NOT NULL,


            FOREIGN KEY (Category_ID)
            REFERENCES Category(Category_ID));
        -- 4. BUDGET TABLE
        CREATE TABLE Budget (
            Budget_ID INTEGER PRIMARY KEY,
            Budget_Amount DECIMAL(10,2) NOT NULL
                CHECK (Budget_Amount > 0),
            Start_Date DATE NOT NULL,
            End_Date DATE NOT NULL,
            Student_ID INTEGER NOT NULL,
            FOREIGN KEY (Student_ID)
            REFERENCES Student(Student_ID));
        -- 5. CART TABLE
        CREATE TABLE Cart (
            Cart_ID INTEGER PRIMARY KEY,
            Created_Date DATE DEFAULT CURRENT_DATE,
            Student_ID INTEGER NOT NULL,
        FOREIGN KEY (Student_ID)
            REFERENCES Student(Student_ID);
        -- 6. CART ITEM TABLE
        CREATE TABLE Cart_Item (
            CartItem_ID INTEGER PRIMARY KEY,
            Quantity INTEGER NOT NULL CHECK (Quantity > 0),
            Product_ID INTEGER NOT NULL,
            Cart_ID INTEGER NOT NULL,
         FOREIGN KEY (Product_ID)
            REFERENCES Product(Product_ID),
          FOREIGN KEY (Cart_ID)
            REFERENCES Cart(Cart_ID)
        );
       -- 7. ORDER DETAILS TABLE
        CREATE TABLE Order_Details (
            Order_ID INTEGER PRIMARY KEY,
            Order_Date DATE DEFAULT CURRENT_DATE,
```

```
    Total_Amount DECIMAL(10,2) NOT NULL
        CHECK (Total_Amount >= 0),
    Student_ID INTEGER NOT NULL,

    FOREIGN KEY (Student_ID)
    REFERENCES Student(Student_ID));
-- ------------------------------------------------------------------------------------------
-- 8. ORDER ITEM TABLE
-- -
CREATE TABLE Order_Item (
    OrderItem_ID INTEGER PRIMARY KEY,
    Quantity INTEGER NOT NULL CHECK (Quantity > 0),
    Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
    Order_ID INTEGER NOT NULL,
    Product_ID INTEGER NOT NULL,
    FOREIGN KEY (Order_ID)
    REFERENCES Order_Details(Order_ID),
-- ------  FOREIGN KEY (Product_ID)
    REFERENCES Product(Product_ID));
-- ------------------------------------------------------------------------------------------
-- 9. PAYMENT TABLE
CREATE TABLE Payment (
    Payment_ID INTEGER PRIMARY KEY,
    Payment_Date DATE DEFAULT CURRENT_DATE,
    Amount DECIMAL(10,2) NOT NULL CHECK (Amount > 0),
    Payment_Method VARCHAR(30) NOT NULL,
    Order_ID INTEGER NOT NULL,
    FOREIGN KEY (Order_ID)
    REFERENCES Order_Details(Order_ID));
-- 10. WISHLIST TABLE
CREATE TABLE Wishlist (
    Wishlist_ID INTEGER PRIMARY KEY,
-- ------------------------------------------------------------------------------------------
    Added_Date DATE DEFAULT CURRENT_DATE,
-- ------------------------------------------------------------------------------------------
    Student_ID INTEGER NOT NULL,
    Product_ID INTEGER NOT NULL,
    FOREIGN KEY (Student_ID)
    REFERENCES Student(Student_ID),
    FOREIGN KEY (Product_ID)
    REFERENCES Product(Product_ID));
```

7.2 Sample Data — DML.

Table	No. of Records	Sample Data / Screenshot Reference
Student	5	dml.sql — 1001, Sruthi, AIML, 2nd Year
Category	10	dml.sql — Electronics, Books, Clothing
Product	12	dml.sql — Laptop Bag, Notebook, Earphones
Budget	15	dml.sql — ₹5,000, ₹3,000, ₹4,500
Cart	5	dml.sql — Cart 101, Cart 102
Payment	50	dml.sql — ₹1,250, UPI

7.3 Constraints & Data Integrity

Constraint / Validation	Applied On	Purpose	Evidence / Screenshot
PRIMARY KEY	Student_ID, Product_ID, Order_ID	Uniquely identifies each record and prevents duplicate key values.	ddl.sql Duplicate primary key is rejected
FOREIGN KEY	Category_ID, Student_ID, Product_ID, Order_ID	Maintains valid relationships between related tables.	ddl.sql Invalid reference is rejected
NOT NULL	Name, Email, Product_Name, Price	Prevents missing values in required attributes.	ddl.sql NULL value is rejected
UNIQUE	Student Email, Coupon Code	Prevents duplicate unique identifiers/contact values.	ddl.sql Duplicate value is rejected
CHECK(Price>0)	Product Price	Ensures product price is valid	ddl.sql Negative/zero price is rejected
DEFAULT	Created_Date, Payment_Date	Automatically stores the current date	ddl.sql Date is inserted automatically

7.4 SQL Operations & Queries

.

S.No	Operation / Query	Purpose	SQL Query / Screenshot Ref.
1	**Create Table**	.Create a new databse table	CREATE TABLE Category (Category_ID INT PRIMARY KEY
2	**Insert Data**	. Add a new product	INSERT INTO Product VALUES (101'Notebook'0;
3	**Update Data**	Update product price.	UPDATE Product SET Price = 75 WHERE Product_ID = 101;

4	**Delete Data**	Delete a product record	DELETE FROM Product WHERE Product_ID = 101;
5	**Select Data**	Display all products	SELECT * FROM Product
6	**WHERE**	. Find products below ₹500	SELECT Product_Name, Price FROM Product WHERE Price < 500;
7	**ORDER BY**	Display products by price	SELECT * FROM Product ORDER BY Price ASC;
8	**JOIN**	Display products with categories	SELECT p.Product_Name, c.Category_Name FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID;

S.No	Operation / Query	Purpose	SQL Query / Screenshot Ref.
9	**COUNT**	. Count total products	SELECT COUNT(*) FROM Product;
10	**SUM**	Calculate total stock	SELECT SUM(Stock) FROM Product;
11	**AVG**	Find average product price	SELECT AVG(Price) FROM Product;
12	**MAX**	Find highest product price	SELECT MAX(Price) FROM Product;
13	**GROUP BY**	Group products by category	SELECT Category_ID, COUNT(*) FROM Product GROUP BY Category_ID;
14	**HAVING**	. Filter categories based on product count	SELECT Category_ID, COUNT(*) FROM Product GROUP BY Category_ID HAVING COUNT(*) > 2;

7.5 Sample Outputs / Screenshots

Figure 1: Display all Students

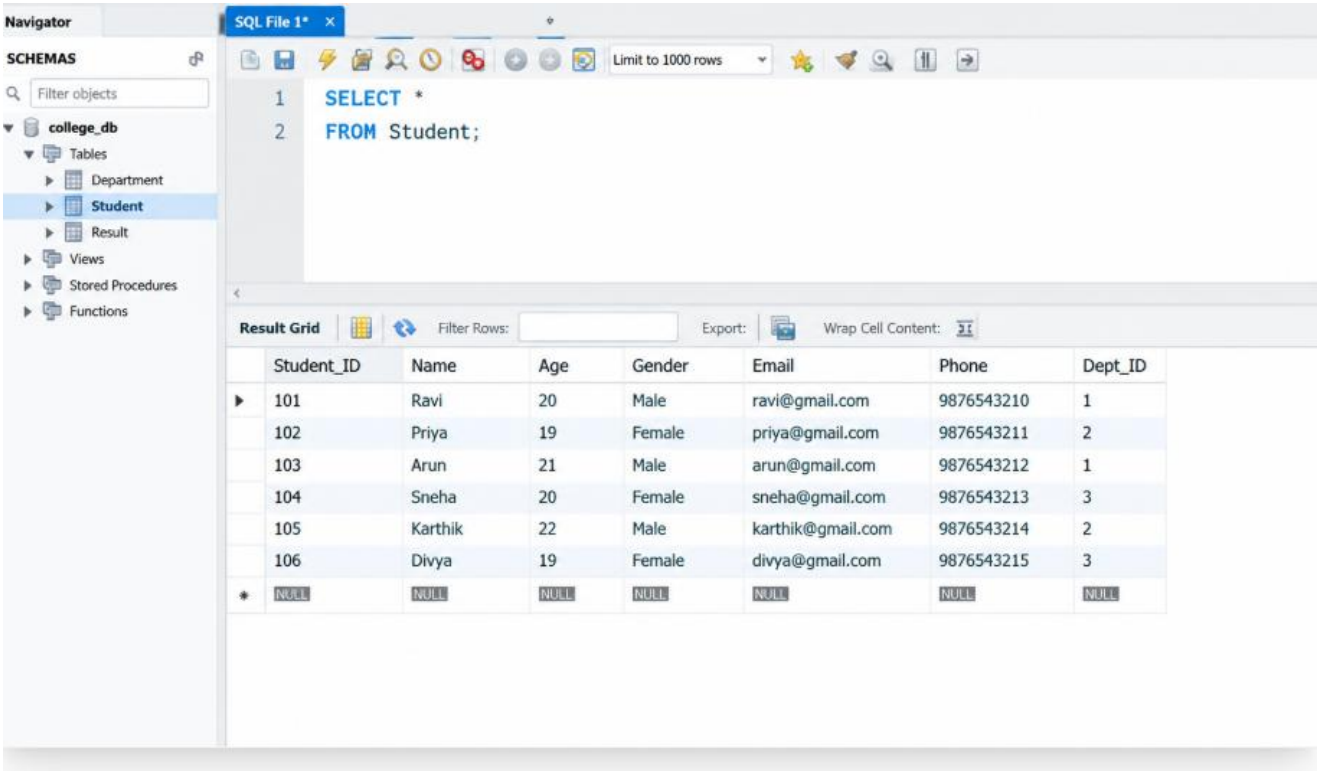


Figure 2: Display all product categories

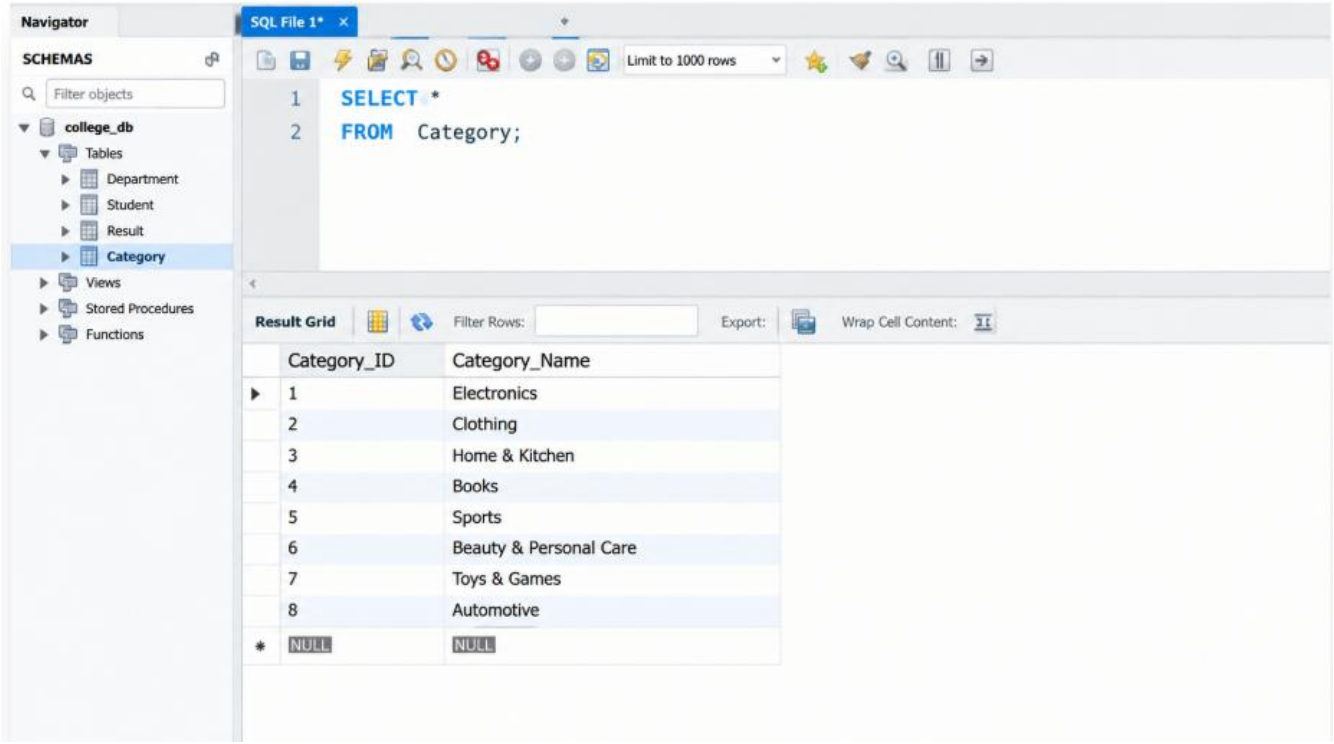


Figure 3 : Display all products

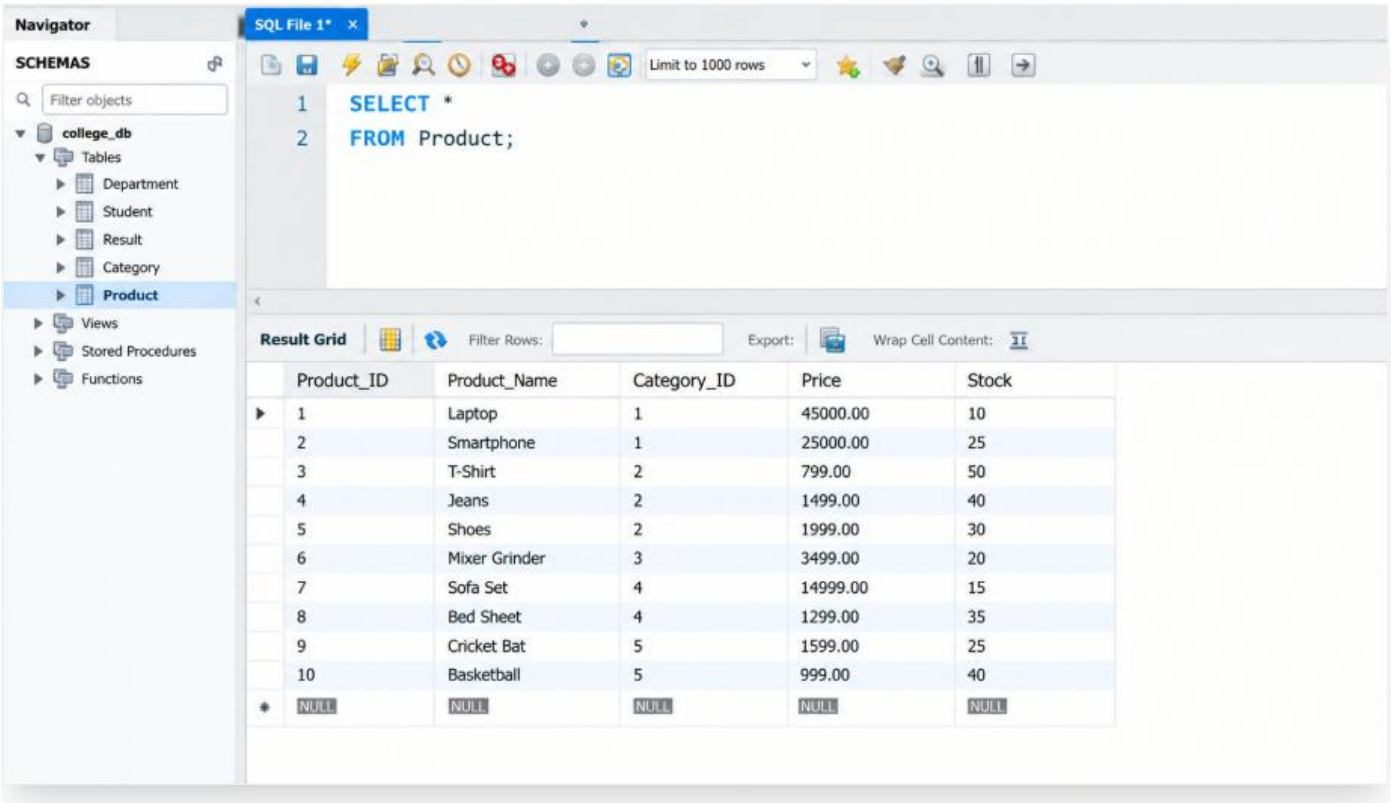


Figure 4: Display all budgets

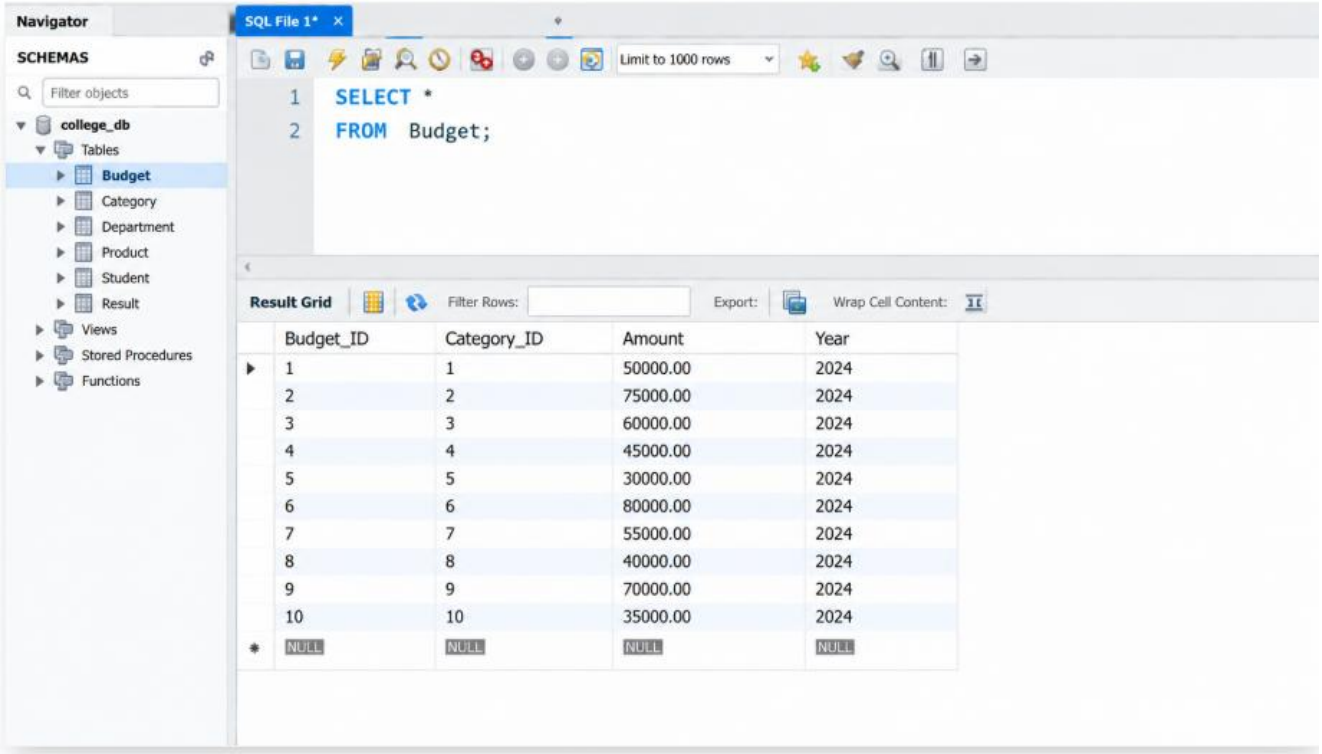


Figure 5: . Products priced below Rs. 5000

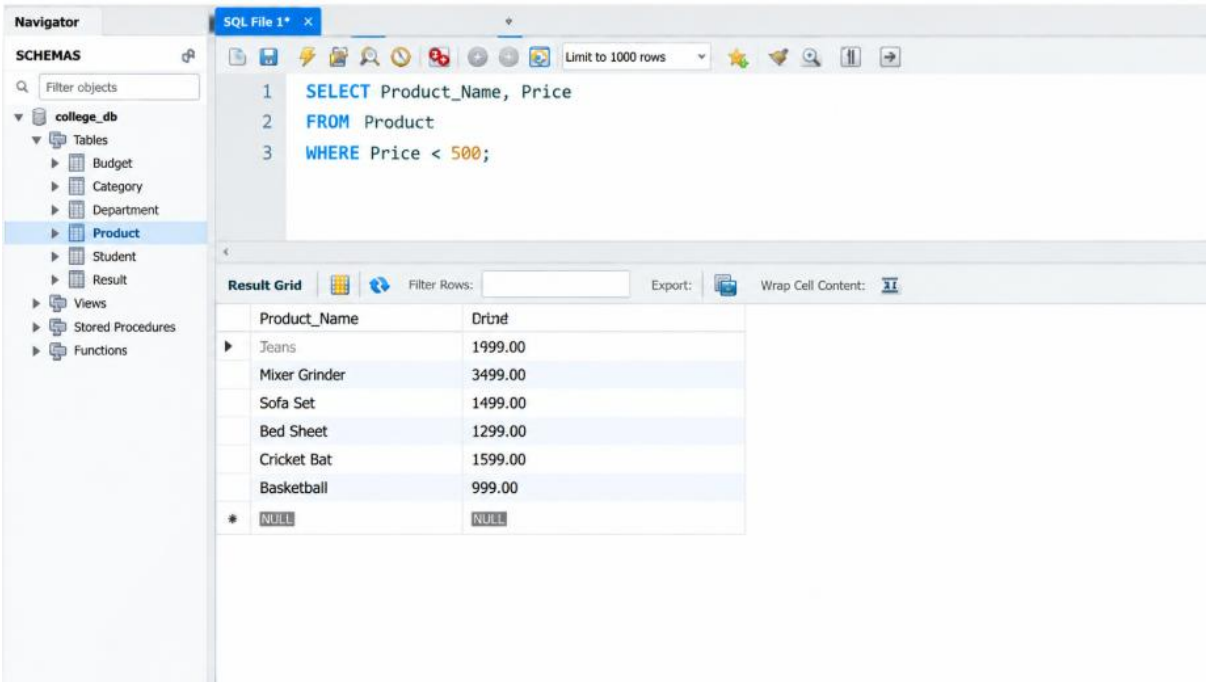
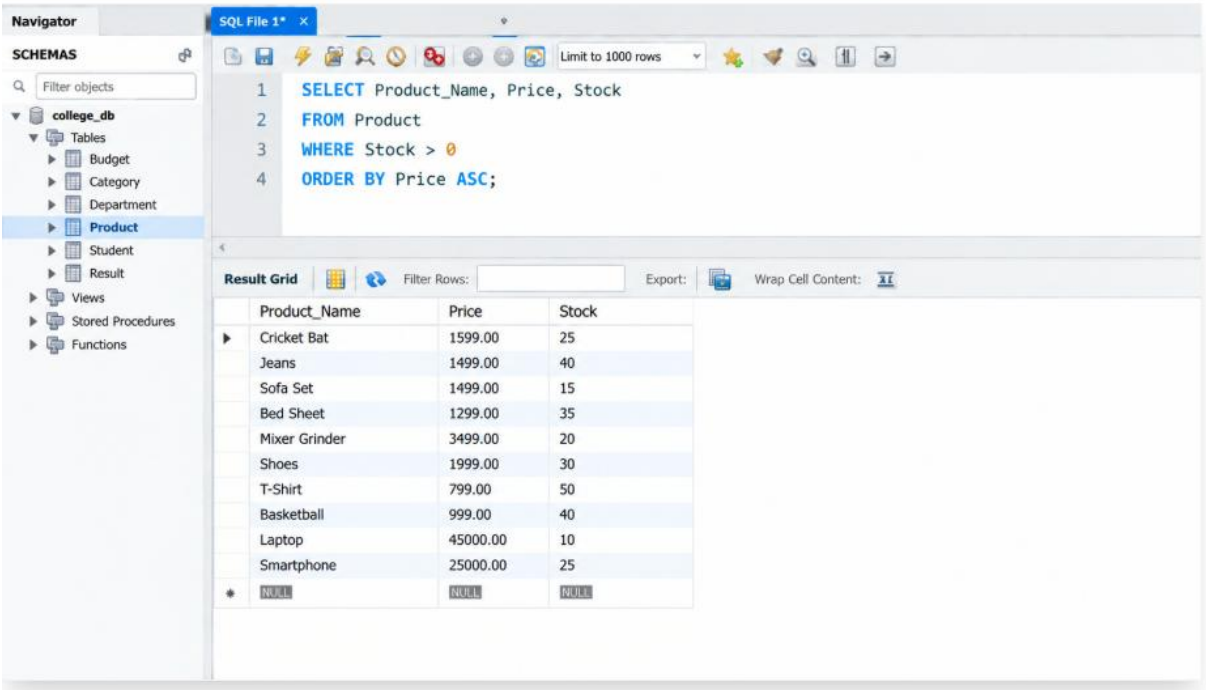


Figure 6: Available products sorted by price (low to high)

```
 SELECT Product_Name, Price, Stock
FROM Product
WHERE Stock > 0
ORDER BY Price ASC;
```

Output:



8 MODULES & FUNCTIONALITIES

S.No	Module Name	Functionality
1	**Student Management**	Stores and manages student details such as name, email, department, and year.
2	**Category Management**	Organizes products into different categori
3	**Product Manageme**	Manages product names, prices, stock, and category details.
4	**Budget Management**	Records and tracks each student's shopping budget.
5	**Cart Management**	Allows students to add and manage products before placing an order
6	**Order Management**	Records orders, order dates, quantities, and total amounts.

9 GITHUB REPOSITORY & TEAM RESPONSIBILITIES

Repository Link: https://github.com/sravani-kotha/student-budget-online-shopping

Student Name	Roll Number	Role / Responsibility	Contribution Status
R.Sruthi Alekhya	25B11AI06	Wrote DDL scripts (tables, keys, constraints)	Completed
Adatra Sumalatha	26B21AI001\	Designed ER diagram & relational schema	Completed
K. Sravani Jyothi	26B21AI059	Wrote CRUD, Join & Aggregate queries	Completed
Y. Y. Manasa Vihari	26B21AI116	Compiled documentation & screenshots	Completed

10.ENHANCEMENTS & FUTURE IMPROVEMENTS

The system can be enhanced by adding AI-based product recommendations, personalized budget suggestions, real-time stock updates, secure online payment, and mobile application support**.** In the future, it can also provide spending analysis and notifications to help students manage their budgets more effectively.

11.CONCLUSION

The **Student Budget Online Shopping System** successfully provides a centralized database for managing student shopping activities in an organized and efficient manner. The system maintains important information such as **student details, product categories, products, budgets, carts, orders, payments, wishlists, feedback, addresses, deliveries, and coupons**. It helps students keep track of their spending and make suitable purchasing decisions within their available budget. The project also demonstrates important **DBMS concepts such as entity relationships, primary keys, foreign keys, constraints, normalization, CRUD operations, joins, aggregate functions, grouping, subqueries, and views**. Proper validation helps maintain data accuracy, consistency, and integrity throughout the system. Overall, the project provides a strong database foundation for a student-friendly online shopping platform and can be further enhanced with **AI recommendations, personalized budget planning, real-time notifications, secure payment integration, and mobile application support** in the future.

12.REFERENCES

1. Uploaded project DDL script: ddl.sql.
2. Uploaded project DML/sample-data script: dml.sql.
3. Uploaded SQL query script: queries.sql.
4. Aditya University — DBMS Capstone Project documentation template.
5. GitHub repository: Bank Account Management System (project repository).

APPENDIX

A. Complete SQL Script

The final SQL implementation is supplied through the uploaded project files:

• ddl.sql — database table definitions, keys, foreign keys, and constraints.

• dml.sql — sample data for all six tables.

• queries.sql — representative SQL operations and analytical queries.

Repository: https://github.com/sravani-kotha/student-budget-online-shopping

Additional Screenshots / Outputs.

```
SELECT s.Student_Name,
    SUM(b.Budget_Amount) AS Total_Budget
FROM Budget b
JOIN Student s ON b.Student_ID = s.Student_ID
GROUP BY s.Student_Name
HAVING SUM(b.Budget_Amount) > 10000
```

